Mukul Prasad

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Education

Ramaiah Institute of Technology, Bengaluru, India — Expected Jun 2028
3rd Year, B.E. in Computer Science & Engineering (AI & ML) — **CGPA: 9.32**
St. Arnold H.S. School, Indore, India — Class XII (2023): **88.0%** | Class X (2021): **95.0%**

Experience

ML Intern | *FlyRank AI* — Jul 2026 – Aug 2026

- Worked on **search intelligence, data contracts, action scoring, and ML experimentation**, developing a **Google Search Ranking & Discoverability** capstone using Python, PyTorch, Scikit-learn, and LLM workflows.

Open Source Mentor | *GirlScript Summer of Code (GSSoC'26)* — Jun 2026 – Aug 2026

- Selected among **135 mentors nationally**, supporting developers across **6 domains**: AI/ML, Web, DevOps/Cloud, Data Science, Security, and Open Source.
- Guided **3 developers** across **5+ engineering areas** including React state management, async workflows, CI automation, debugging, and code review; accumulated **60 mentor points** through hands-on open-source mentoring.

Research Intern, AI/ML Drug Repurposing | *IEEE Computer Society IAMPro* — Apr 2026 – Jun 2026

- Built an AI drug-repurposing pipeline spanning **1,500+ drugs, 20K+ genes, 130+ diseases, and 50K+ edges**, using PyTorch Geometric, RDKit, PubMed, WHO data, and **2,048-bit Morgan fingerprints**.
- Co-authoring an **IEEE research paper** by developing the **4-component NDPI** framework combining HGT predictions, chemical similarity, PubMed evidence, and WHO equity scoring.

Projects

NeuroSOC — Neuromorphic AI Security Operations Center — *1st Runner-up — CyreneAI Hackathon*
Python, PyTorch, SNNs, LNNs, XGBoost, FastAPI, Kafka, React — GitHub | Demo

- Engineered a neuromorphic threat detection platform combining SNN, LNN, and XGBoost, achieving **99.6% accuracy** and **0.961 macro F1** on **83,325** public-dataset test rows; built an **80-feature** telemetry pipeline, real-time Kafka streaming, autonomous honeypot sandboxing & validated **19/19** browser end-to-end steps.

Jvalyx — Satellite-Based Industrial Fire Intelligence — *Semifinalist — Smart India Hackathon 2026*
Python, FastAPI, CatBoost, Isolation Forest, GeoPandas, React, WebSockets — GitHub | Demo

- Designed a geospatial hazard intelligence pipeline for **5 fire-event classes**, combining satellite detections, temporal baselines, sensor fusion, and anomaly scoring into explainable **0–100** risk scores, with interactive incident replay, real-time event updates, evidence cards, and counterfactual escalation from routine flare to critical industrial hazard.

Project Astraeus — AI Mission Control — *SIH 2025 Merit*
Python, TensorFlow, GNN, PPO, RL, Digital Twin Architecture — Github | Demo | Live

- Engineered an AI-powered digital twin for satellite mission control featuring real-time telemetry, intelligent communication scheduling using **GNN + PPO**, and **91% predictive failure detection accuracy**.

RegGraph-AI — Regulatory Graph Intelligence — *Zonal Champion, Finalist — HackArena 2.0, IIIT Delhi*
Python, Graph AI, NLP, Deep Learning, RAG — Github | Demo

- Built a Graph AI pipeline for regulatory intelligence using knowledge graphs, NLP entity extraction, and RAG, delivering explainable compliance retrieval by surfacing regulatory risk pathways from large document corpora.

Technical skills

Languages: Python, C,C++, Java, SQL
AIML: PyTorch, Scikit-learn, HF, LangChain, LangGraph, NLP, RAG Pipelines, Multi-Agent Systems, Deep Learning
Web & Backend: Node.js, MERN Stack, REST APIs, Firebase
Tools: Git, GitHub(2.6k+ contributions), MCP(Integration and Operations)
Cloud & MLOps: AWS (Lambda, S3, SageMaker, EC2, IAM), Docker, CI/CD, Model Deployment
Databases: PostgreSQL, MongoDB, MySQL, Redis, Supabase, Hive, Hbase, Graph DB(Neo4j)
Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems,Complier Design

Achievements

- **ML Competitions: Top 150/10,760** (as of Jul 2026) - Kaggle Wellbore Geology Prediction(International team led by PyJa) | Top 25/348 - Kaggle Formula 1 Race Result Classification (Polaris Institute of Technology)
- **Hackathons:** National Semifinalist @Flipkart Grid 8.0, National Semifinalist@ JP Morgan Chase-Code for Good'26| 1st runner-up -GenAI Fusion'25 Hack,Async by CyreneAI, TCS HackDay | Zonal Champion- HackArena 2.0 (IIIT Delhi) | Top 10/1,000 - BIT MESRA Hack'24 | Finalist - Samsung PRISM'25,'26, wHACKiest'26, Hack-A-War'25 (Top10), Codecrafter'26, HackMalenadu'26 | SIH'26 SemiFinalist
- **Academic:** Nearly perfect **9.95 SGPA** (Sem II) | 9.61 SGPA (Sem III) - Ramaiah Institute of Technology
- **Certifications:** AWS Cloud Practitioner Essentials | AWS Fundamentals of AI/ML | IBM India Impact GenAI Program - IISc Bengaluru | AI Fluency: Framework & Foundations by Anthropic
- **Leadership & POR:** Leading **MSRIT'28** & **MSRIT'29** communities (**3.5K+ active members**) | Founding Core Member- Amazon Web Services (AWS) Cloud Club | Core Member -Quantum Cluster(QpiAI Industry MoU) | Webmaster- IEEE CIS | **DeepStation (US) Hack Organizer-Google for Dev** | E-Cell|DSC(formerly GDG)